Coolant Level Signal for LLOYD's Malfunction

Symptoms

- The OEM signal for LLOYD's sensor has malfunctioned.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

This LLOYD's sensor is connected to the OEM side (X7 connector) of the remote input/output unit. The OEM is responsible for this connection.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the OEM wiring harness.	
	STEP 1A. Check the coolant level signal and sensor supply +24-VDC wires for an open.	
	STEP 1B. Check the coolant level signal and sensor supply +24-VDC wires for a wire-to-wire short.	
	STEP 1C. Check the coolant level signal wire for a short to ground.	
	STEP 1D. Check the coolant level sensor supply +24-VDC for voltage.	

STEP 1. Check the OEM wiring harness.

STEP 1A. Check the coolant level signal and sensor supply +24-VDC wires for an open.

Conditions :

- Disconnect the OEM harness at the X7 connector.
- Disconnect the coolant level sensor connector.

Action	Specification/Repair	Next Step
Check the coolant level signal and sensor supply +24-VDC wires for an open. • Place one test lead on the coolant level signal pin at the X7 connector. • Place the other test lead on the coolant level signal pin at the sensor connector. • Place one test lead on the coolant level sensor supply +24-VDC pin at the X7 connector. • Place the other test lead on the coolant level sensor supply +24-VDC pin at the sensor connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2B**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to the OEM service manual.	**Repair complete**

STEP 2B. Check the coolant level signal and sensor supply +24-VDC wires for a wire-to-wire short.

Conditions :

- Disconnect the OEM harness at the X7 connector.
- Disconnect the coolant level sensor connector.

Action	Specification/Repair	Next Step

Check the coolant level signal and sensor supply +24-VDC wires for a wire-to-wire short. • Place one test lead on the coolant level signal pin at the X7 connector. • Place the other test lead on all other pins at the X7 connector. • Place one test lead on the coolant level sensor supply +24-VDC pin at the X7 connector. • Place the other test lead on all other pins at the X7 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the OEM service manual.	**Repair complete**
	Less than 10 ohms? **NO**	**2C**

STEP 2C. Check the coolant level signal and sensor supply +24-VDC wires for a short to ground.

Conditions :

- Disconnect the OEM harness at the X7 connector.
- Disconnect the coolant level sensor connector.

Action	Specification/Repair	Next Step

Check the coolant level signal and sensor supply +24-VDC wires for a short to ground. - Place one test lead on the coolant level signal pin at the X7 connector. - Place the other test lead on engine ground. - Place one test lead on the sensor supply +24-VDC pin at the X7 connector. - Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the OEM service manual.	**Repair complete**
	Less than 10 ohms? **NO**	**2D**

STEP 2D. Check the coolant level sensor supply +24-VDC wire for voltage.

Conditions :

- Disconnect the coolant level sensor connector.

Action	Specification/Repair	Next Step

Check the coolant level sensor supply +24-VDC wire for voltage. • Place one test lead on the coolant level sensor supply +24-VDC pin at the sensor connector. • Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	+24-VDC? **YES** **Repair :** Replace the coolant level sensor. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	**Repair complete**
	+24-VDC? **NO** **Repair :** Check the batteries. Refer to the OEM service manual. Replace the remote input/output unit. Contact a Cummins® Authorized Repair Location.	**Repair complete**

Last Modified: 11-Jul-2008